

A Large-Target Row Embedding of S_2^m into S_p^n for All p

Run `fa_banach_001`, lane 1

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Status. Full solution, likely valid, for the large-target existence reading. This note records a simple positive embedding for the $q = 2$ case left open in the introduction of Chattopadhyay–Hong–Pal–Pradhan–Ray. It is not a minimal-dimension classification.

1 Source question

The source paper is:

A. Chattopadhyay, G. Hong, A. Pal, C. Pradhan, and S. K. Ray,

Isometric embeddability of S_q^m into S_p^n , arXiv:2008.13164v4, Journal of Functional Analysis 282 (2022), article 109281, DOI 10.1016/j.jfa.2021.109281.

On page 2, after Theorem 1.1, the authors state that they do not know whether S_2^m embeds isometrically into S_p^n for $1 < p < \infty$ when p is not an even integer. They compare this with the even- p case, where known results imply embeddings for sufficiently large n , and they also mention the unresolved cases $S_q^m \rightarrow S_1^n$ for $q \in \{2, 3\}$.

implicitly proved in [13] that S_1^m does not embed isometrically into S_∞^n . Their proof relied on a clever trick based on unitary dilation of contractions on a Hilbert space. Later, Pisier guessed that the same result is true when S_∞^n is replaced by S_∞ (see [13]) and G. Misra asked whether ℓ_q^m embeds into S_∞^n for all $1 \leq q \neq 2 < \infty$ and $2 \leq m \leq n$. All these questions were settled in [40] using Birkhoff-James orthogonality and norm parallelism, combined with infinite descent type arguments. It was also proved in [40] that S_1^m does not embed isometrically into S_p for all $m \geq 2$ and $1 < p < \infty$. However, the question, whether ℓ_q , $q \in (1, 2)$ embeds isometrically into S_∞^n was left open in [40]. Thus in Theorem 1.1, we settle this remaining case, by giving yet another proof which is completely different from [40] and new even in the commutative case. However, we do not know if S_2^m embeds isometrically into S_p^n where $1 < p < \infty$ and p is not an even integer. If $p \in 2\mathbb{N}$ and $m \in \mathbb{N}$ are given, then it can be easily seen from [32, Theorem B] that S_2^m embeds isometrically into S_p^n for any $n \in \mathbb{N}$, which is sufficiently large. Moreover, we also do not know whether S_q^m embeds isometrically into S_1^n , where $n \geq m \geq 2$ and $q \in \{2, 3\}$.

Let us briefly mention some key steps and new ingredients used in the proof of Theorem 1.1 when $1 < n \neq q < \infty$

2 Definitions

For $1 \leq p < \infty$, S_p^n denotes $M_n(\mathbb{C})$ with norm

$$\|X\|_{S_p^n} = \left(\sum_j s_j(X)^p \right)^{1/p},$$

where $s_j(X)$ are the singular values of X . For $p = \infty$, S_∞^n is $M_n(\mathbb{C})$ with the operator norm. The space S_2^m is the Hilbert space $M_m(\mathbb{C})$ with Hilbert–Schmidt norm

$$\|A\|_{S_2^m} = (\text{Tr}(A^*A))^{1/2}.$$

3 Proof Intuition

The source paper already observes that S_2^2 embeds into S_∞^n for $n \geq 4$ through a row or column Hilbert space. The same row idea works for every Schatten exponent p , not only $p = \infty$. A one-row matrix has rank at most one, so there is only one singular value to measure. Every Schatten norm, finite or infinite, therefore returns exactly that singular value. Since S_2^m is a Hilbert space of dimension m^2 , it can be placed in one row of M_n once $n \geq m^2$.

Theorem 1 (large-target row embedding). *Let $m \geq 1$, let $n \geq m^2$, and let $1 \leq p \leq \infty$. Then there exists a complex-linear isometric embedding*

$$\Phi : S_2^m \longrightarrow S_p^n.$$

Proof. Let $\text{vec} : M_m(\mathbb{C}) \rightarrow \mathbb{C}^{m^2}$ list the entries of a matrix in any fixed order. Then

$$\|\text{vec}(A)\|_{\ell_2^{m^2}} = \|A\|_{S_2^m}.$$

Since $n \geq m^2$, define $r(A) \in \mathbb{C}^n$ by padding $\text{vec}(A)$ with zeros. Let $\Phi(A) \in M_n(\mathbb{C})$ be the matrix whose first row is $r(A)$ and whose remaining rows are zero.

The map $A \mapsto \Phi(A)$ is complex-linear. Also,

$$\Phi(A)\Phi(A)^* = \begin{pmatrix} \|r(A)\|_2^2 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Thus $\Phi(A)$ has singular values

$$\|r(A)\|_2, 0, \dots, 0.$$

For $1 \leq p < \infty$,

$$\|\Phi(A)\|_{S_p^n} = (\|r(A)\|_2^p)^{1/p} = \|r(A)\|_2 = \|A\|_{S_2^m}.$$

For $p = \infty$, the operator norm of $\Phi(A)$ is again its only nonzero singular value, namely $\|r(A)\|_2 = \|A\|_{S_2^m}$. Hence Φ is an isometry. In particular it is injective. $\square$

Corollary 1. *For every $m \geq 2$ and every non-even $1 < p < \infty$, S_2^m embeds linearly isometrically into S_p^n for every $n \geq m^2$.*

Proof. Apply the theorem. $\square$

Corollary 2. *For every $m \geq 2$, S_2^m embeds linearly isometrically into S_1^n for every $n \geq m^2$.*

Proof. Apply the theorem with $p = 1$. $\square$

4 Scope and limitations

This result is intentionally limited to large target dimension. It does not decide the minimal n , nor does it solve the $q = 3$ case for $S_q^m \rightarrow S_1^n$. It also does not assert complete isometry, preservation of self-adjointness, trace preservation, multiplicativity, or compatibility with the matrix algebra structure. The construction is a plain Banach-space linear isometry, matching the stated convention of the source paper’s finite-dimensional theorem.

5 Verification and novelty check

The proof uses only the singular-value calculation for a one-row matrix. The optional script `code/verify_row_embedding.py` numerically checks random examples.

Before promotion, the run’s cheap indexes were searched for arXiv:2008.13164 and core terms including `S_2m`, `S_pn`, Schatten, row embedding, and isometric embeddability. Nearby indexed packets concern different Schatten questions or the later $p < 1$ quasi-Banach paper arXiv:2207.09062. Web searches for the exact source title and the non-even- p phrase found the source publication and the later quasi-Banach extension, but no exact large-target row-embedding packet in this run.

Novelty confidence as a global literature claim is modest, because row Hilbert subspaces of Schatten classes are standard. Mathematical validity confidence is high. Human review should decide whether the source sentence was meant literally as a large-target existence question or instead as shorthand for a small- n , optimal-dimension, self-adjoint, algebraic, or complete-isometric variant. Under the large-target reading, the answer is complete: $n = m^2$ suffices for all p .

6 References

- [1] A. Chattopadhyay, G. Hong, A. Pal, C. Pradhan, and S. K. Ray, *Isometric embeddability of S_q^m into S_p^n* , J. Funct. Anal. 282 (2022), no. 1, Paper No. 109281. arXiv:2008.13164.
- [2] A. Chattopadhyay, G. Hong, C. Pradhan, and S. K. Ray, *On isometric embeddability of S_q^m into S_p^n as non-commutative quasi-Banach spaces*, arXiv:2207.09062; Proc. Roy. Soc. Edinburgh Sect. A, published online 2023.